The density matrix: relaxation and optical response

Week

IV

In Week III we followed the evolution of a closed two-level atom. A coherent optical field creates a superposition of the two states, and the expectation value of the electric dipole gives the polarization that enters Maxwell's equations. Away from resonance, perturbation theory gave a linear susceptibility with the frequency dependence of an undamped Lorentz oscillator. Near resonance, the interaction picture and rotating-wave approximation allowed us to obtain bounded Rabi oscillations. Yet, **absorption is missing, which is connected with the missing of a description of relaxation**.

Empirically, we know that the excited state has a finite lifetime, and the relative phase of a superposition can be disturbed by its environment. An ensemble may also contain atoms prepared in different states. A single state vector is then insufficient to describe the atomic state available to us.

To bridge this gap, this week introduces the density operator, derives its equation of motion, and identifies the different roles of population decay and dephasing. We will finish by deriving the two-level susceptibility, including its linewidth and saturation, and explaining precisely when it agrees and disagrees with the Lorentz model, as well as additional advantages of the description.

Remembering the conventions carried from Week III: The ordered basis is $\{|1\rangle, |2\rangle\}$, with $|1\rangle$ lower in energy. We use

$$\omega_0 = \frac{E_2 - E_1}{\hbar} > 0, \qquad \Delta = \omega_0 - \omega, \qquad E(t) = E_0 \cos \omega t. \tag{0.1}$$

The real number E_0 is the peak electric field. For a fixed linear polarization, choose the relative phase of the states so that the projected transition dipole is $d = p_{12} = p_{21} > 0$. Then

$$\hat{p} = d(\hat{\sigma}_- + \hat{\sigma}_+), \qquad \Omega = \frac{dE_0}{\hbar}, \tag{0.2}$$

where $\hat{\sigma}_- = |1\rangle\langle 2|$, $\hat{\sigma}_+ = |2\rangle\langle 1|$, and $\hat{\sigma}_z = |2\rangle\langle 2| - |1\rangle\langle 1|$.

1 From state vectors to the density operator

Let us begin with a pure state written in a known atomic basis,

$$|\psi(t)\rangle = \sum_n c_n(t)|n\rangle. \tag{1.1}$$

where the coefficients include the full Schrödinger-picture time dependence. An expectation value is

$$\langle \hat{O} \rangle = \sum_{m,n} c_m^* c_n \langle m|\hat{O}|n\rangle, \tag{1.2}$$

meaning that observables depend on products of amplitudes, rather than on an individual amplitude alone. This motivates the definition of the density operator for a pure state.

Definition. Density operator of a pure state

$$\boxed{\hat{\rho} = |\psi\rangle\langle\psi|, \qquad \rho_{mn} = \langle m|\hat{\rho}|n\rangle = c_m c_n^*.} \tag{1.3}$$

For a two-level atom,

$$\rho = \begin{pmatrix} |c_1|^2 & c_1 c_2^* \\ c_2 c_1^* & |c_2|^2 \end{pmatrix} = \begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix}. \tag{1.4}$$

It is trivial to interpret each term: the diagonal elements can be interpreted as the probabilities of finding the atom in the corresponding basis states, which we call **populations**; the off-diagonal elements retain information about relative phases between basis components, and are called **coherences**.

Note that multiplying $|\psi\rangle$ by a common phase leaves $\hat{\rho}$ unchanged, but a relative phase generally changes the off-diagonal elements. The density operator therefore discards the unobservable overall phase while retaining the phase information needed for interference.

1.1 Statistical mixtures and what the probabilities mean

While we described the density operator of a pure state, the density operator is also the way to describe an ensemble (e.g. of atoms) whose members need not have the same pure state. Suppose preparation a occurs with probability P_a, producing the normalized state $|\psi_a\rangle$. Then

Definition. Density operator of a statistical ensemble.

$$\boxed{\hat{\rho} = \sum_a P_a |\psi_a\rangle\langle\psi_a|, \qquad P_a \geq 0, \quad \sum_a P_a = 1.} \tag{1.5}$$

Writing $|\psi_a\rangle = \sum_n c_n^{(a)} |n\rangle$ gives

$$\rho_{mn} = \sum_a P_a c_m^{(a)} c_n^{(a)*}. \tag{1.6}$$

Note that the preparation states **need not be orthogonal**, nor P_a correspond to a population ρ_{aa}. Note also that the same density operator can arise from different preparation procedures. For example,

$$\frac{1}{2}|1\rangle\langle 1| + \frac{1}{2}|2\rangle\langle 2| = \frac{1}{2}|+\rangle\langle +| + \frac{1}{2}|-\rangle\langle -|, \tag{1.7}$$

where $|\pm\rangle = (|1\rangle \pm |2\rangle)/\sqrt{2}$. These ensembles give identical results for every measurement on the atom. In this sense, the density operator specifies measurement statistics but does not uniquely identify how the ensemble was prepared.

Observation: A superposition is different from a mixture. Compare the coherent state

$$|\psi_\phi\rangle = \frac{|1\rangle + e^{i\phi}|2\rangle}{\sqrt{2}}, \qquad \rho_{\text{coh}} = \frac{1}{2}\begin{pmatrix} 1 & e^{-i\phi} \\ e^{i\phi} & 1 \end{pmatrix}, \tag{1.8}$$

with an equal statistical mixture of the two levels,

$$\rho_{\text{mix}} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{1.9}$$

Both have $\rho_{11} = \rho_{22} = 1/2$. Energy measurements alone cannot distinguish them. However, their dipoles are

$$\langle\hat{p}\rangle_{\text{coh}} = d\cos\phi, \qquad \langle\hat{p}\rangle_{\text{mix}} = 0. \tag{1.10}$$

Even when $\cos\phi = 0$, another phase-sensitive measurement distinguishes the states. For example, projection onto $|\psi_\phi\rangle$ succeeds with probability one in the coherent state and one half in the mixture.

1.2 Expectation values and physical constraints

For an ensemble, the quantum expectation must be averaged over preparations, which means that

$$\begin{aligned}\langle\hat{O}\rangle &= \sum_a P_a \langle\psi_a|\hat{O}|\psi_a\rangle \\ &= \sum_{m,n} \rho_{mn} O_{nm} = \boxed{\mathrm{Tr}\big(\hat{\rho}\hat{O}\big)}.\end{aligned} \tag{1.11}$$

Note the reversed indices in $\rho_{mn}O_{nm}$. For example if $\hat{\sigma}_{ij} = |i\rangle\langle j|$, one has $\langle\hat{\sigma}_{ij}\rangle = \rho_{ji}$. In particular,

$$\boxed{\langle\hat{p}\rangle = d(\rho_{21} + \rho_{12}) = 2d\,\mathrm{Re}(\rho_{21}).} \tag{1.12}$$

Density Operator properties:
A physical density operator satisfies

- $\hat{\rho}^\dagger = \hat{\rho}$, from Eq. 1.5;
- $\mathrm{Tr}\,\hat{\rho} = 1$, from normalization;
- $\langle\varphi|\hat{\rho}|\varphi\rangle \geq 0$ for every $|\varphi\rangle$., from the fact that each term $P_a|\langle\varphi|\psi_a\rangle|^2$ is nonnegative.
- For a normalized pure state, $\hat{\rho}^2 = \hat{\rho}$, and its purity is $\mathrm{Tr}\,\hat{\rho}^2 = 1$.
- If λ_j are the eigenvalues of a density operator in a Hilbert space of dimension D, then

$$\frac{1}{D} \leq \mathrm{Tr}\,\hat{\rho}^2 = \sum_j \lambda_j^2 \leq 1. \tag{1.13}$$

  The upper limit occurs for a pure state, while the lower limit occurs for the maximally mixed state $\hat{\mathbb{I}}/D$.

For two-level systems, these conditions imply

$$\rho_{11} + \rho_{22} = 1, \qquad \rho_{12} = \rho_{21}^*, \qquad \boxed{|\rho_{21}|^2 \leq \rho_{11}\rho_{22}.} \tag{1.14}$$

The last inequality follows from the non-negative determinant, which is a useful check that states that a model cannot produce an arbitrarily large coherence at fixed populations. Besides, a coherent two-level superposition has purity one, whereas the equal mixture has purity one half.

The normalized $\hat{\rho}$ thus describes one atom, or the statistics of a represen-

tative atom in an ensemble. For identical independent atoms experiencing the same local conditions we have that the polarization shall be given by

$$\boxed{\boldsymbol{P}(t) = \mathcal{N}\ \mathrm{Tr}(\hat{\rho}(t)\hat{\boldsymbol{p}}).} \tag{1.15}$$

The number density $\mathcal{N}$ supplies the number of dipoles per unit volume[1].

[1] *it is not part of the trace-one normalization of ρ!*

Note that a complete many-atom density operator still lives in a tensor-product Hilbert space, but is unnecessary for this independent-atom, one-body polarization. This means that energies and transition matrix elements can be enough to construct this model, without obtaining every electronic wavefunction explicitly.

2 Time-evolution of the density operator

For a closed system, the state vector obeys the Schrödinger equation,

$$|\dot{\psi}_a\rangle = -\frac{i}{\hbar}\hat{H}|\psi_a\rangle, \qquad \langle\dot{\psi}_a| = \frac{i}{\hbar}\langle\psi_a|\hat{H}. \tag{2.1}$$

where the second equation follows by Hermitian conjugation. For fixed preparation probabilities P_a, differentiating Eq. 1.5 gives

$$\dot{\hat{\rho}} = \sum_a P_a \left(|\dot{\psi}_a\rangle\langle\psi_a| + |\psi_a\rangle\langle\dot{\psi}_a|\right) = -\frac{i}{\hbar}\hat{H}\hat{\rho} + \frac{i}{\hbar}\hat{\rho}\hat{H}.$$

which can be recast into the

Liouville–von Neumann equation:

$$\boxed{\frac{d\hat{\rho}}{dt} = -\frac{i}{\hbar}[\hat{H}, \hat{\rho}].} \tag{2.2}$$

An important observation is that while it resembles an Heisenberg equation, this equation applies to pure states in Schrödinger-picture or statistical mixtures evolving under an Hamiltonian that may depend on time. In a fixed complete basis, this yields a set of coupled equations[2]

[2] *for now on we will use the dot notation for simplicity.*

$$\boxed{\dot{\rho}_{mn} = -\frac{i}{\hbar}\sum_k \left(H_{mk}\rho_{kn} - \rho_{mk}H_{kn}\right).} \tag{2.3}$$

2.1 Unitary evolution and free coherence

For free evolution, $H_{mn} = E_n\delta_{mn}$, so

$$\dot{\rho}_{mn} = -i\omega_{mn}\rho_{mn}, \qquad \rho_{mn}(t) = \rho_{mn}(0)e^{-i\omega_{mn}t}. \tag{2.4}$$

which means that the populations remain fixed, whereas coherences oscillate with the Bohr frequency[3]. In a two-level atom, the optical coherence obeys $\rho_{21}(t) = \rho_{21}(0)e^{-i\omega_0 t}$, giving precisely the freely oscillating dipole obtained from the state vector previously. A diagonal mixture remains diagonal and has zero mean dipole.

[3] $\omega_{mn} = \omega_m - \omega_n$

Also, using the evolution operator, it can be shown that the eigenvalues of $\hat{\rho}$, and hence its purity, are unchanged by closed-system evolution. Directly,

$$\frac{d}{dt}\operatorname{Tr}\hat{\rho} = 0, \qquad \frac{d}{dt}\operatorname{Tr}\hat{\rho}^2 = -\frac{2i}{\hbar}\operatorname{Tr}\Big(\hat{\rho}[\hat{H},\hat{\rho}]\Big) = 0. \tag{2.5}$$

where the last equation uses the cyclic property of the trace that makes the commutator terms cancel.

3 Relaxation processes and the master equation

The atom may interact with degrees of freedom that we do not observe, such as vacuum radiation modes or a surrounding material. The combined atom-environment state can evolve unitarily while the atomic state alone does not. We obtain that state by a partial trace,

$$\hat{\rho}_A = \operatorname{Tr}_B \hat{\rho}_{AB} = \sum_b \langle b|\hat{\rho}_{AB}|b\rangle. \tag{3.1}$$

Here A denotes the atom and B its environment. If $\hat{H}_{AB} = \hat{H}_A + \hat{H}_B + \hat{H}^{\text{int}}_{AB}$, then tracing the full equation of motion gives the exact relation

$$\dot{\hat{\rho}}_A = -\frac{i}{\hbar}[\hat{H}_A,\hat{\rho}_A] - \frac{i}{\hbar}\operatorname{Tr}_B[\hat{H}^{\text{int}}_{AB},\hat{\rho}_{AB}]. \tag{3.2}$$

The final term depends on the joint state, which means that this form is not yet a closed equation for the atom. Approximations about the reservoir are needed to replace it by a relaxation superoperator acting on $\hat{\rho}_A$ alone.

3.1 A controlled form for the relaxation terms

For a weakly coupled, rapidly decorrelating reservoir, a Markov description is often appropriate, which means assuming that the coarse-grained evolution depends on the present state rather than its history. With the usual secular approximation, a physically consistent generator for the relaxation processes has the Lindblad form

$$\boxed{\dot{\hat{\rho}} = -\frac{i}{\hbar}[\hat{H},\hat{\rho}] + \sum_\alpha \mathcal{D}[\hat{L}_\alpha]\hat{\rho},} \tag{3.3}$$

where

$$\mathcal{D}[\hat{L}]\hat{\rho} = \hat{L}\hat{\rho}\hat{L}^\dagger - \frac{1}{2}\left(\hat{L}^\dagger\hat{L}\hat{\rho} + \hat{\rho}\hat{L}^\dagger\hat{L}\right). \tag{3.4}$$

The jump operators $\hat{L}_\alpha$ specify the relaxation channels, with rates absorbed into their amplitudes. **For nonnegative rates, this form preserves the trace and generates completely positive evolution.** Complete positivity means that the evolution remains positive even if the atom is correlated with an untouched auxiliary system. Trace preservation alone is not sufficient to guarantee it.

3.2 Spontaneous emission: population loss and repopulation

The first process to model is the spontaneous emission. Consider decay from $|2\rangle$ into $|1\rangle$ at rate Γ. The jump operator is

$$\hat{L}_\Gamma = \sqrt{\Gamma}\,\hat{\sigma}_-, \qquad \mathcal{L}_{\text{decay}}(\hat{\rho}) = \Gamma\mathcal{D}[\hat{\sigma}_-]\hat{\rho}. \tag{3.5}$$

Using the superoperator defined above, i.e., $\mathcal{D}[\hat{L}]\hat{\rho}$ it is possible to obtain[4]

[4] *a trivial linear algebra exercise*

$$\boxed{\mathcal{L}_{\text{decay}}(\rho) = \Gamma\begin{pmatrix} \rho_{22} & -\rho_{12}/2 \\ -\rho_{21}/2 & -\rho_{22} \end{pmatrix}.} \tag{3.6}$$

This matrix means that the excited population decays at rate Γ, and exactly the same population enters the lower state. The optical coherence decays at rate $\Gamma/2$ from this process alone, and a direct consequence of the dissipator, not an independently chosen damping constant.

The two-level manifold is thus closed to population loss - every atom remains in one of its two levels - but still considered an open quantum system because it exchanges energy and correlations with the radiation field.

3.3 Pure dephasing without population transfer

Other physical processes may fluctuate the transition phase without transferring population. A convenient two-level model uses

$$\hat{L}_\phi = \sqrt{\frac{\gamma_\phi}{2}}\,\hat{\sigma}_z. \tag{3.7}$$

Since $\hat{\sigma}_z^2 = \hat{\mathbb{I}}$,

$$\begin{aligned}\mathcal{L}_{\text{deph}}(\rho) &= \frac{\gamma_\phi}{2}(\hat{\sigma}_z\rho\hat{\sigma}_z - \rho) \\ &= \begin{pmatrix} 0 & -\gamma_\phi\rho_{12} \\ -\gamma_\phi\rho_{21} & 0 \end{pmatrix}.\end{aligned} \tag{3.8}$$

Thus γ_ϕ is defined to be the additional coherence decay rate. The coefficient of $\mathcal{D}[\sigma_z]$ is $\gamma_\phi/2$, not γ_ϕ.

3.4 Incoherent pumping and the relaxation times

An incoherent upward transition $|1\rangle \to |2\rangle$ at rate R is described by $\hat{L}_R = \sqrt{R}\,\hat{\sigma}_+$. Its contribution is

$$\mathcal{L}_{\text{pump}}(\rho) = R \begin{pmatrix} -\rho_{11} & -\rho_{12}/2 \\ -\rho_{21}/2 & \rho_{11} \end{pmatrix}. \tag{3.9}$$

It transfers population upward and also damps coherence. It does not supply a fixed phase, unlike a coherent laser drive. This pump acts within the two retained levels and preserves their total population; injection or removal of atoms would require a separate number-density model.

Combining the three channels, our master equation is

$$\boxed{\begin{aligned} \dot{\hat{\rho}} = & -\frac{i}{\hbar}[\hat{H}, \hat{\rho}] + \Gamma\mathcal{D}[\hat{\sigma}_-]\hat{\rho} \\ & + \frac{\gamma_\phi}{2}\mathcal{D}[\hat{\sigma}_z]\hat{\rho} + R\mathcal{D}[\hat{\sigma}_+]\hat{\rho}. \end{aligned}} \tag{3.10}$$

Define the ground-minus-excited population difference $w = \rho_{11} - \rho_{22}$ and

$$\gamma_1 = \Gamma + R, \qquad w_{\text{eq}} = \frac{\Gamma - R}{\Gamma + R}, \qquad \gamma_2 = \frac{\Gamma + R}{2} + \gamma_\phi. \tag{3.11}$$

For $\Gamma + R > 0$, in the absence of the coherent drive,

$$\dot{w} = -\gamma_1(w - w_{\text{eq}}), \qquad \dot{\rho}_{21} = -(\gamma_2 + i\omega_0)\rho_{21}. \tag{3.12}$$

The population and coherence relaxation times are therefore

$$\boxed{T_1 = \frac{1}{\gamma_1}, \qquad T_2 = \frac{1}{\gamma_2}, \qquad \frac{1}{T_2} = \frac{1}{2T_1} + \gamma_\phi.} \tag{3.13}$$

In this model $T_2 \le 2T_1$, with equality if there is no pure dephasing. We will set $R = 0$ for the passive optical response below, so $w_{\text{eq}} = 1$, $\gamma_1 = \Gamma$, and $\gamma_2 = \Gamma/2 + \gamma_\phi$.

3.5 Undriven decay and amplitude damping

With $R = 0$ and the laser off, the solution is

$$\boxed{\begin{aligned}\rho_{22}(t) &= \rho_{22}(0)e^{-\Gamma t},\\ \rho_{11}(t) &= 1 - \rho_{22}(0)e^{-\Gamma t},\\ \rho_{21}(t) &= \rho_{21}(0)e^{-(\gamma_2 + i\omega_0)t}.\end{aligned}} \tag{3.14}$$

An initially excited atom has $\rho_{22}(t) = e^{-\Gamma t}$ and zero mean dipole throughout this undriven decay. Yet it emits spontaneously: a zero mean dipole does not imply zero emission intensity, which involves field fluctuations and dipole correlations.

Its atomic purity is $e^{-2\Gamma t} + (1 - e^{-\Gamma t})^2$. It begins at one, decreases while atom and radiation become correlated, and returns to one as the atom reaches the pure ground state. Relaxation does not necessarily decrease purity monotonically. In contrast, pure dephasing of an equal superposition leaves the populations at one half and gives purity $(1 + e^{-2\gamma_\phi t})/2$ when $\Gamma = 0$.

Replacing E_2 by $E_2 - i\hbar\Gamma/2$ in a state-vector equation gives the effective Hamiltonian

$$\hat{H}_{\text{eff}} = \hat{H} - \frac{i\hbar\Gamma}{2}|2\rangle\langle 2|. \tag{3.15}$$

It correctly describes an unnormalized state conditioned on no emission event. By itself it loses norm and omits the jump term that repopulates $|1\rangle$. It is therefore not the unconditional master equation for the ensemble. This is why adding decay to the Week III amplitudes is insufficient for the present problem.

4 The driven two-level atom

4.1 From the commutator to population and coherence equations

We now apply the general master equation to the atom studied in Week III. Set the lower-state energy to zero, which removes only a common energy. The Schrödinger-picture Hamiltonian is

$$\hat{H}(t) = \hbar\omega_0|2\rangle\langle 2| - dE(t)(\hat{\sigma}_- + \hat{\sigma}_+), \tag{4.1}$$

or, in our ordered basis,

$$H = \begin{pmatrix} 0 & -dE(t) \\ -dE(t) & \hbar\omega_0 \end{pmatrix}. \tag{4.2}$$

Keep $R = 0$. To illustrate the component calculation explicitly,

$$[H,\rho]_{22} = H_{21}\rho_{12} - \rho_{21}H_{12} = -dE(t)(\rho_{12} - \rho_{21}), \tag{4.3}$$

$$\begin{aligned}[H,\rho]_{21} &= H_{21}\rho_{11} + H_{22}\rho_{21} - \rho_{21}H_{11} - \rho_{22}H_{21} \\ &= \hbar\omega_0\rho_{21} - dE(t)(\rho_{11} - \rho_{22}).\end{aligned} \tag{4.4}$$

Adding the relaxation terms already derived gives

$$\boxed{\begin{aligned}\dot\rho_{22} &= -\Gamma\rho_{22} + \frac{idE(t)}{\hbar}(\rho_{12} - \rho_{21}), \\ \dot\rho_{11} &= +\Gamma\rho_{22} - \frac{idE(t)}{\hbar}(\rho_{12} - \rho_{21}), \\ \dot\rho_{21} &= -(\gamma_2 + i\omega_0)\rho_{21} + \frac{idE(t)}{\hbar}(\rho_{11} - \rho_{22}), \\ \dot\rho_{12} &= \dot\rho_{21}^*.\end{aligned}} \tag{4.5}$$

The first two equations conserve $\rho_{11} + \rho_{22}$. The coherent field transfers population through the coherence, while spontaneous emission transfers excited population into the lower state. No rotating-wave approximation has yet been made in these equations.

4.2 RWA and a rotating density matrix

The steps now parallel thesis Secs. 3.2.1–3.2.2 and the interaction-picture analysis of Week III. For $E(t) = E_0\cos\omega t$, the interaction picture separates terms at $\omega_0 - \omega$ from terms at $\omega_0 + \omega$. In the regime

$$|\Delta|, \Omega, \Gamma, \gamma_\phi \ll \omega_0 + \omega, \tag{4.6}$$

we discard the counter-rotating terms. The Schrödinger-picture Hamiltonian becomes

$$\hat H_{\rm RWA}(t) = \hbar\omega_0|2\rangle\langle 2| - \frac{\hbar\Omega}{2}\left(e^{-i\omega t}\hat\sigma_+ + e^{i\omega t}\hat\sigma_-\right). \tag{4.7}$$

Introduce the rotating-frame transformation

$$\hat R(t) = |1\rangle\langle 1| + e^{i\omega t}|2\rangle\langle 2|, \qquad \widetilde\rho = \hat R\hat\rho\hat R^\dagger. \tag{4.8}$$

Differentiating this expression and using the original equation of motion shows that the transformed Hamiltonian is

$$\hat H_{\rm rot} = \hat R\hat H_{\rm RWA}\hat R^\dagger + i\hbar\dot{\hat R}\hat R^\dagger. \tag{4.9}$$

The second term is $-\hbar\omega|2\rangle\langle 2|$, and the phases in the coupling cancel. Hence

$$\boxed{\hat{H}_{\rm rot} = \hbar\Delta|2\rangle\langle 2| - \frac{\hbar\Omega}{2}(\hat{\sigma}_+ + \hat{\sigma}_-).} \tag{4.10}$$

Subtracting $\hbar\Delta\hat{\mathbb{I}}/2$ recovers the midpoint form used in Week III. The rotating-frame change itself is exact for the RWA Hamiltonian; the approximation was the removal of the fast coupling terms.

The jump operators must also be transformed. Here $\hat{R}\hat{\sigma}_-\hat{R}^\dagger = e^{-i\omega t}\hat{\sigma}_-$ and $\hat{R}\hat{\sigma}_z\hat{R}^\dagger = \hat{\sigma}_z$. Since $\mathcal{D}[e^{i\theta}L] = \mathcal{D}[L]$, the dissipators keep the same form. The populations are unchanged, whereas the coherences become

$$\widetilde{\rho}_{11} = \rho_{11}, \qquad \widetilde{\rho}_{22} = \rho_{22}, \qquad \widetilde{\rho}_{21} = e^{i\omega t}\rho_{21}, \qquad \widetilde{\rho}_{12} = \widetilde{\rho}_{21}^*. \tag{4.11}$$

The rotating coherence removes the optical carrier from the matrix element we need to calculate. We retain ρ_{11} and ρ_{22} for the populations, since they are the same in both frames.

4.3 Equations for the density-matrix elements

Using Eq. 4.10 in the master equation, we obtain

$$\boxed{\begin{aligned}
\dot{\rho}_{22} &= -\Gamma\rho_{22} + \frac{i\Omega}{2}(\widetilde{\rho}_{12} - \widetilde{\rho}_{21}),\\
\dot{\rho}_{11} &= +\Gamma\rho_{22} - \frac{i\Omega}{2}(\widetilde{\rho}_{12} - \widetilde{\rho}_{21}),\\
\dot{\widetilde{\rho}}_{21} &= -(\gamma_2 + i\Delta)\widetilde{\rho}_{21} + \frac{i\Omega}{2}(\rho_{11} - \rho_{22}),\\
\dot{\widetilde{\rho}}_{12} &= -(\gamma_2 - i\Delta)\widetilde{\rho}_{12} - \frac{i\Omega}{2}(\rho_{11} - \rho_{22}).
\end{aligned}} \tag{4.12}$$

These equations describe the populations and coherences directly. Hermiticity gives $\widetilde{\rho}_{12} = \widetilde{\rho}_{21}^*$, while trace preservation gives $\rho_{11} = 1 - \rho_{22}$. Thus only one population and one complex coherence need to be solved independently. In particular,

$$\frac{i\Omega}{2}(\widetilde{\rho}_{12} - \widetilde{\rho}_{21}) = \Omega\,\mathrm{Im}\,\widetilde{\rho}_{21}, \tag{4.13}$$

so the imaginary part of the coherence determines the coherent transfer of population. Conversely, the difference $\rho_{11} - \rho_{22}$ drives the coherence. The sign of the detuning term follows from $\Delta = \omega_0 - \omega$ and from using the matrix element $\widetilde{\rho}_{21}$.

Population and coherence relaxation have separate rates; no assumption $T_1 = T_2$ is needed. The coherent terms are not a perturbation expansion in the field strength. Within the validity of the Hamiltonian and relaxation

model, these equations describe weak excitation and saturation alike.

The density matrix is

$$\widetilde{\rho} = \begin{pmatrix} \rho_{11} & \widetilde{\rho}_{12} \\ \widetilde{\rho}_{21} & \rho_{22} \end{pmatrix}. \tag{4.14}$$

Its physical constraints can also be stated directly in terms of its elements:

$$\rho_{11}, \rho_{22} \geq 0, \qquad \rho_{11} + \rho_{22} = 1, \qquad |\widetilde{\rho}_{21}|^2 \leq \rho_{11}\rho_{22}. \tag{4.15}$$

The last inequality follows from the nonnegative determinant. Its purity is

$$\mathrm{Tr}\left(\widetilde{\rho}^2\right) = \rho_{11}^2 + \rho_{22}^2 + 2|\widetilde{\rho}_{21}|^2. \tag{4.16}$$

Without relaxation, unitary evolution preserves this purity and reproduces the coherent dynamics of Week III. With spontaneous emission and no drive or pump, the state approaches $\rho_{11} = 1$, $\rho_{22} = 0$, and $\widetilde{\rho}_{21} = 0$.

If an incoherent pump is retained, the independent equations become

$$\begin{aligned} \dot{\rho}_{22} &= -\Gamma\rho_{22} + R\rho_{11} + \frac{i\Omega}{2}(\widetilde{\rho}_{12} - \widetilde{\rho}_{21}), \\ \dot{\widetilde{\rho}}_{21} &= -\left(\frac{\Gamma + R}{2} + \gamma_\phi + i\Delta\right)\widetilde{\rho}_{21} + \frac{i\Omega}{2}(\rho_{11} - \rho_{22}). \end{aligned} \tag{4.17}$$

The remaining elements follow from normalization and Hermiticity. This is a useful extension, but the following passive response has $R = 0$.

4.4 How relaxation changes Rabi oscillations

At resonance, $\Delta = 0$, an atom initially in $|1\rangle$ has $\rho_{11}(0) = 1$, $\rho_{22}(0) = 0$, and $\widetilde{\rho}_{21}(0) = 0$. The real part of the coherence obeys

$$\frac{d}{dt}\,\mathrm{Re}\,\widetilde{\rho}_{21} = -\gamma_2\,\mathrm{Re}\,\widetilde{\rho}_{21}, \tag{4.18}$$

and therefore remains zero. Using $\rho_{11} = 1 - \rho_{22}$, the remaining equations are

$$\begin{aligned} \dot{\rho}_{22} &= -\Gamma\rho_{22} + \Omega\,\mathrm{Im}\,\widetilde{\rho}_{21}, \\ \frac{d}{dt}\,\mathrm{Im}\,\widetilde{\rho}_{21} &= -\gamma_2\,\mathrm{Im}\,\widetilde{\rho}_{21} + \frac{\Omega}{2}(1 - 2\rho_{22}). \end{aligned} \tag{4.19}$$

Differentiate the population equation and substitute the coherence equation:

$$\ddot{\rho}_{22} = -\Gamma\dot{\rho}_{22} - \Omega\gamma_2\,\mathrm{Im}\,\widetilde{\rho}_{21} + \frac{\Omega^2}{2}(1 - 2\rho_{22}). \tag{4.20}$$

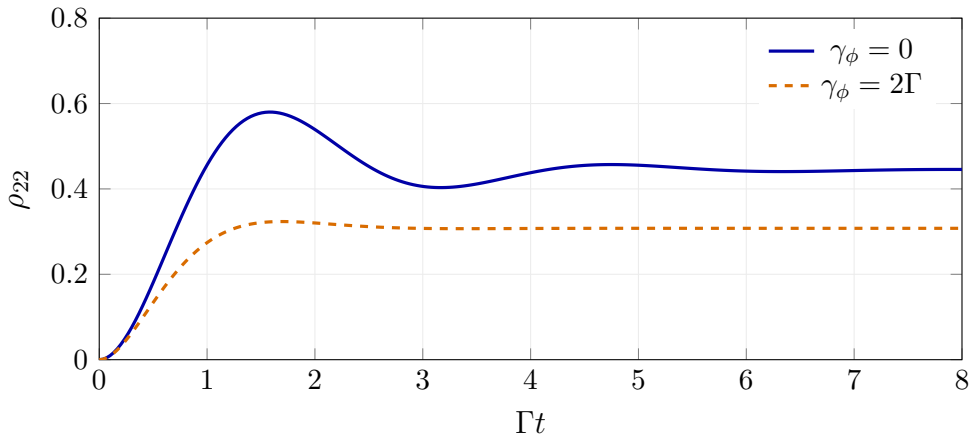


Figure 1. Resonant excitation from the ground state with $\Omega = 2\Gamma$ and no incoherent pump. Both curves follow Eq. 4.23. Additional pure dephasing damps the oscillations and, for this fixed drive, lowers the steady excited population.

Using $\Omega \operatorname{Im} \widetilde{\rho}_{21} = \dot{\rho}_{22} + \Gamma \rho_{22}$ eliminates the coherence and gives

$$\ddot{\rho}_{22} + (\Gamma + \gamma_2)\dot{\rho}_{22} + (\Gamma\gamma_2 + \Omega^2)\rho_{22} = \frac{\Omega^2}{2}. \tag{4.21}$$

For the underdamped case, define

$$a = \frac{\Gamma + \gamma_2}{2}, \qquad \nu = \sqrt{\Omega^2 - \frac{(\Gamma - \gamma_2)^2}{4}}. \tag{4.22}$$

The initial conditions $\rho_{22}(0) = \dot{\rho}_{22}(0) = 0$ give

$$\rho_{22}(t) = \frac{\Omega^2}{2(\Gamma\gamma_2 + \Omega^2)} \left[1 - e^{-at}\left(\cos \nu t + \frac{a}{\nu} \sin \nu t\right)\right]. \tag{4.23}$$

When $\nu^2 < 0$, the corresponding hyperbolic functions describe an overdamped approach. At critical damping the bracketed transient is obtained by the continuous limit $\nu \to 0$. The system can show damped Rabi oscillations, but it eventually reaches a stationary rotating-frame state. This long-time state was absent in our lossless model.

5 From coherence to optical susceptibility

5.1 The polarization envelope and its normalization

The connection to the macroscopic field follows the thesis, Sec. 3.2.3. From Eq. 1.15,

$$P(t) = \mathcal{N} d \left[\widetilde{\rho}_{21}(t) e^{-i\omega t} + \widetilde{\rho}_{12}(t) e^{i\omega t}\right]. \tag{5.1}$$

Define the complex envelopes by

$$E(t) = \mathrm{Re}[E_0 e^{-i\omega t}], \qquad P(t) = \mathrm{Re}[\widetilde{P}(t) e^{-i\omega t}]. \tag{5.2}$$

Term-by-term comparison gives

$$\boxed{\widetilde{P} = 2\mathcal{N} d \widetilde{\rho}_{21}.} \tag{5.3}$$

The factor of two arises because the positive-frequency coefficient of a real field of peak amplitude E_0 is $E_0/2$. It cannot be omitted when matching the coherence to the susceptibility. In a rotating-frame steady state,

$$\widetilde{P} = \epsilon_0 \chi E_0, \qquad \boxed{\chi = \frac{2\mathcal{N} d}{\epsilon_0 E_0} \widetilde{\rho}_{21}^{\mathrm{ss}} = \frac{2\mathcal{N} d^2}{\epsilon_0 \hbar \Omega} \widetilde{\rho}_{21}^{\mathrm{ss}}.} \tag{5.4}$$

At finite intensity this defines an effective susceptibility at the drive frequency. Only its weak-field limit is the intensity-independent linear susceptibility.

For a transverse field in a nonmagnetic medium, the polarization can serve as the source in the scalar wave equation,

$$\nabla^2 E - \frac{1}{c^2} \frac{\partial^2 E}{\partial t^2} = \mu_0 \frac{\partial^2 P}{\partial t^2}. \tag{5.5}$$

Together with the internal-state master equation, this is the local basis of the Maxwell–Bloch description in thesis Sec. 2.3. We do not solve propagation here: the field at the atom is prescribed, and we calculate the constitutive response $P(E)$.

5.2 Linear response in the stationary limit

We will now consider a case where populations and coherences have already stabilized. If we set all derivatives in Eq. 4.12 to zero, the coherence equation gives

$$\widetilde{\rho}_{21}^{\mathrm{ss}} = \frac{\Omega}{2} (\rho_{11}^{\mathrm{ss}} - \rho_{22}^{\mathrm{ss}}) \frac{\Delta + i\gamma_2}{\Delta^2 + \gamma_2^2}. \tag{5.6}$$

Since $\rho_{11}^{\mathrm{ss}} = 1 - \rho_{22}^{\mathrm{ss}}$, the population equation becomes

$$\Gamma \rho_{22}^{\mathrm{ss}} = \Omega \, \mathrm{Im}\, \widetilde{\rho}_{21}^{\mathrm{ss}} = \frac{\Omega^2 \gamma_2}{2(\Delta^2 + \gamma_2^2)} (1 - 2\rho_{22}^{\mathrm{ss}}). \tag{5.7}$$

For $\Gamma > 0$, define the saturation parameter

$$s(\Delta) = \frac{\Omega^2 \gamma_2}{\Gamma(\Delta^2 + \gamma_2^2)}, \qquad s_0 = s(0) = \frac{\Omega^2}{\Gamma \gamma_2}. \tag{5.8}$$

The population equation then reads $\rho_{22}^{\mathrm{ss}} = s(1 - 2\rho_{22}^{\mathrm{ss}})/2$. Solving it and substituting into the coherence equation gives

$$\boxed{\begin{aligned} \rho_{22}^{\mathrm{ss}} &= \frac{s}{2(1+s)}, \qquad \rho_{11}^{\mathrm{ss}} = \frac{2+s}{2(1+s)}, \\ \widetilde{\rho}_{21}^{\mathrm{ss}} &= \frac{\Omega}{2(1+s)} \frac{\Delta + i\gamma_2}{\Delta^2 + \gamma_2^2}, \qquad \widetilde{\rho}_{12}^{\mathrm{ss}} = (\widetilde{\rho}_{21}^{\mathrm{ss}})^*. \end{aligned}} \tag{5.9}$$

In particular, $\rho_{11}^{\mathrm{ss}} - \rho_{22}^{\mathrm{ss}} = 1/(1+s)$. For $s \ll 1$, the lower state is nearly fully populated and linear response is justified. For $s \gg 1$, the populations approach one half. A single coherent drive does not produce stationary population inversion in this unpumped two-level model, even though a coherent pulse can temporarily transfer all the population upward.

Let $A = \mathcal{N}d^2/(\epsilon_0\hbar)$, which has units of angular frequency. Using Eq. 5.4,

$$\boxed{\chi(\omega, E_0) = \frac{\chi^{(1)}(\omega)}{1+s} = A \frac{\Delta + i\gamma_2}{\Delta^2 + \gamma_2^2 + \Omega^2\gamma_2/\Gamma}.} \tag{5.10}$$

Although the master equation is linear in ρ for a prescribed field, its stationary solution depends nonlinearly on that field. The resulting polarization saturates because the drive reduces the population difference that creates the coherence. Equation 5.10 is a field-dependent response, not a new linear susceptibility.

For a plane wave in vacuum, $I = \epsilon_0 c E_0^2/2$. Defining

$$I_{\mathrm{sat}} = \frac{\epsilon_0 c \hbar^2 \Gamma \gamma_2}{2d^2} \tag{5.11}$$

gives $s_0 = I/I_{\mathrm{sat}}$ and $s(\Delta) = s_0/[1 + (\Delta/\gamma_2)^2]$. At fixed incident intensity the absorption line has full width

$$\boxed{\delta\omega_{\mathrm{FWHM}} = 2\gamma_2\sqrt{1+s_0},} \tag{5.12}$$

while its resonant height decreases by $1/(1+s_0)$. This is power broadening together with saturation. An orientationally distributed ensemble requires averaging the intensity-dependent response over the projected couplings; replacing d^2 by one average inside s is not generally sufficient.

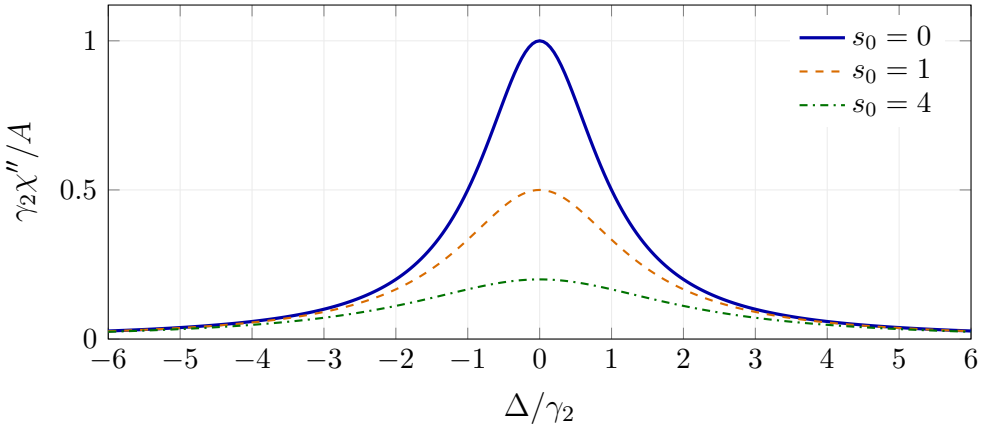


Figure 2. Absorptive response from Eq. 5.10, plotted against detuning at fixed intensity for each curve. The weak-field case is a Lorentzian with full width $2\gamma_2$. Increasing intensity lowers its peak and broadens it. Here $s_0 = 0$ denotes the linear-response limit.

5.3 Comparison with the Lorentz oscillator

The classical Lorentz model describes an electron of mass m_e bound by a harmonic restoring force. Its susceptibility is

$$\chi_{\mathrm{L}}(\omega) = \frac{\mathcal{N}e^2}{\epsilon_0 m_e}\frac{1}{\omega_{\mathrm{L}}^2 - \omega^2 - i\gamma_{\mathrm{L}}\omega}, \tag{5.13}$$

where γ_{L} multiplies the velocity in the oscillator equation. To compare with this linear model, we must first take the weak-field limit of the quantum response, rather than compare it with a saturated atom.

For $s \ll 1$, $\rho_{11}^{\mathrm{ss}} \simeq 1$ and $\rho_{22}^{\mathrm{ss}} \simeq 0$. The stationary coherence and the resulting susceptibility are

$$\tilde{\rho}_{21}^{(1),\mathrm{ss}} = \frac{i\Omega}{2(\gamma_2 + i\Delta)} = \frac{\Omega}{2}\frac{\Delta + i\gamma_2}{\Delta^2 + \gamma_2^2}, \tag{5.14}$$

$$\boxed{\chi^{(1)}(\omega) = \frac{\mathcal{N}d^2}{\epsilon_0\hbar}\frac{1}{\Delta - i\gamma_2} = A\frac{\Delta + i\gamma_2}{\Delta^2 + \gamma_2^2}.} \tag{5.15}$$

Here $\Delta = \omega_0 - \omega$, as throughout these notes. The response depends on the dipole matrix element of the selected transition and on the decay rate of its coherence.

To compare the denominators, set $\omega_{\mathrm{L}} = \omega_0$ and consider a narrow interval around resonance, $|\Delta|, \gamma_2 \ll \omega_0$. Then

$$\omega_0^2 - \omega^2 = (\omega_0 - \omega)(\omega_0 + \omega) \simeq 2\omega_0\Delta, \qquad \gamma_{\mathrm{L}}\omega \simeq \gamma_{\mathrm{L}}\omega_0, \tag{5.16}$$

so that

$$\chi_{\mathrm{L}}(\omega) \simeq \frac{\mathcal{N}e^2}{2\epsilon_0 m_e \omega_0} \frac{1}{\Delta - i\gamma_{\mathrm{L}}/2}. \tag{5.17}$$

The line shapes therefore agree near resonance when

$$\boxed{\gamma_{\mathrm{L}} = 2\gamma_2 = \frac{2}{T_2}, \qquad f_{12} = \frac{2m_e\omega_0 d^2}{\hbar e^2}, \qquad \chi^{(1)} \simeq f_{12}\chi_{\mathrm{L}}.} \tag{5.18}$$

The dimensionless f_{12} is the oscillator strength for this particular transition and polarization. A single transition need not have unit oscillator strength; the comparison uses the same number density and multiplies the classical spectral weight by f_{12}. A tabulated oscillator strength summed over unresolved magnetic sublevels need not equal the value for one selected pair of states.

The amplitude of a weakly damped classical oscillator decays at $\gamma_{\mathrm{L}}/2$, whereas the quantum coherence decays at γ_2. This explains the factor of two. If spontaneous emission is the only relaxation channel, $\gamma_2 = \Gamma/2$, and $\gamma_{\mathrm{L}} = \Gamma$. With pure dephasing, instead, $\gamma_{\mathrm{L}} = \Gamma + 2\gamma_\phi$: the optical linewidth no longer determines the population lifetime T_1 by itself.

Dispersion and absorption. Writing $\chi^{(1)} = \chi' + i\chi''$ gives

$$\boxed{\chi' = A\frac{\Delta}{\Delta^2 + \gamma_2^2}, \qquad \chi'' = A\frac{\gamma_2}{\Delta^2 + \gamma_2^2}.} \tag{5.19}$$

For our $e^{-i\omega t}$ convention, $\chi' > 0$ below resonance and $\chi' < 0$ above it. The imaginary part is positive for a passive ground-state ensemble, and has full width at half maximum

$$\delta\omega_{\mathrm{FWHM}} = 2\gamma_2 = \Gamma + 2\gamma_\phi. \tag{5.20}$$

This is an angular-frequency width; the width in hertz is γ_2/π. At finite intensity, the real and imaginary parts follow from Eq. 5.10 by replacing their common denominator by $\Delta^2 + \gamma_2^2 + \Omega^2\gamma_2/\Gamma$.

For the full classical expression, put $C = \mathcal{N}e^2/(\epsilon_0 m_e)$. Multiplying numerator and denominator by the complex conjugate yields

$$\mathrm{Re}(f_{12}\chi_{\mathrm{L}}) = f_{12}C\frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2)^2 + \gamma_{\mathrm{L}}^2\omega^2}, \tag{5.21}$$

$$\mathrm{Im}(f_{12}\chi_{\mathrm{L}}) = f_{12}C\frac{\gamma_{\mathrm{L}}\omega}{(\omega_0^2 - \omega^2)^2 + \gamma_{\mathrm{L}}^2\omega^2}. \tag{5.22}$$

These are the expressions compared with the quantum result in Fig. 3. The curves agree near resonance; the small differences arise because the classical

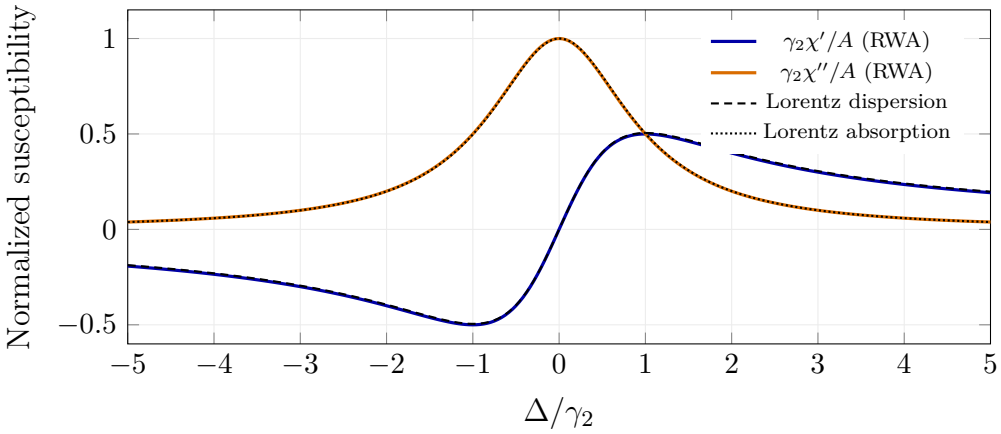


Figure 3. Weak-field quantum and classical responses near resonance. The Lorentz curves plot $f_{12}\chi_{\mathrm{L}}$ with $\omega_{\mathrm{L}} = \omega_0 = 100\gamma_2$ and $\gamma_{\mathrm{L}} = 2\gamma_2$, normalized by A/γ_2. They nearly overlap the RWA quantum result; the small differences reflect the near-resonance approximation.

expression retains its full frequency dependence while Eq. 5.15 uses the RWA.

The absorption interpretation follows from the cycle-averaged work done on the polarization:

$$\left\langle E\frac{dP}{dt}\right\rangle_{\text{cycle}} = \frac{\omega\epsilon_0 E_0^2}{2}\chi''. \tag{5.23}$$

A positive χ'' means that energy is removed from the driving mode. If relaxation is radiative, that energy can subsequently leave as fluorescence. For a dilute medium, $n \simeq 1 + \chi'/2$ and the intensity attenuation coefficient is $\alpha \simeq (\omega/c)\chi''$.

5.4 Comment on the classical correspondence

The linear oscillator reproduces the dispersive and absorptive line shapes because a weakly excited two-level coherence obeys a damped linear equation. This agreement is strongest when one transition dominates, the field is weak, the linewidth and detuning are small compared with the optical frequency, and atoms can be treated independently.

The density-matrix description also identifies what the phenomenological classical parameters represent. It distinguishes population decay from pure dephasing, predicts saturation through the finite population difference, and describes coherent transients and mixed states. A strictly harmonic Lorentz oscillator with fixed parameters has no upper bound on its excitation and does not produce the factor $1/(1+s)$ or power broadening by itself.

With an incoherent pump, the weak-field RWA result is proportional to the undriven population difference $\rho_{11}^{\mathrm{eq}} - \rho_{22}^{\mathrm{eq}} = (\Gamma - R)/(\Gamma + R)$ and uses the pump-modified γ_2. A negative population difference then gives gain rather

than absorption. The passive ground-state Lorentz model has no such population variable. This illustrates why agreement of the weak-field spectra does not make the underlying quantum and classical models equivalent in every regime.

Additional practice exercises

Exercise 1. Compare $|+\rangle\langle+|$, $|1\rangle\langle 1|$, and $\hat{\mathbb{I}}/2$. Find their populations, coherences, eigenvalues, and purities. Which pairs can be distinguished by an energy measurement? By a projection onto $|+\rangle$?

Exercise 2. For $\rho = \left(\begin{smallmatrix} 1-p & z^* \\ z & p \end{smallmatrix}\right)$, determine the allowed region of p and z from positivity. Find the maximum possible magnitude of the mean dipole and identify a state attaining it.

Exercise 3. Derive every entry of Eq. 3.6 by explicit matrix multiplication. Verify trace preservation. Then show why including only the negative excited-population term gives an incomplete model for a closed two-level manifold.

Exercise 4. For $\Gamma^{-1} = 20\,\mathrm{ns}$ and $\gamma_\phi = 0$, calculate T_1, T_2, and the weak-field full linewidth in megahertz. Repeat for $\gamma_\phi = \Gamma/2$. State which lifetime has changed.

Exercise 5. Derive the Hamiltonian in the rotating frame. Show that the decay dissipator is invariant under the phase acquired by σ_-. Recover the equations for ρ_{11}, ρ_{22}, $\widetilde{\rho}_{21}$, and $\widetilde{\rho}_{12}$. Verify trace preservation and preservation of Hermiticity.

Exercise 6. Starting from the stationary density-matrix equations, derive ρ_{11}^{ss}, ρ_{22}^{ss}, $\widetilde{\rho}_{21}^{\mathrm{ss}}$, and $\chi(\omega, E_0)$. Show that on resonance, without pure dephasing, the linear-response condition is $2\Omega^2/\Gamma^2 \ll 1$.